

DAILY INTERNSHIP REPORT

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SECTION 1 — STUDENT INFORMATION

Student Name	P Koteswara rao	VTU ID	VTU 28129
Branch / Dept.	B.Tech / CSE(DS)	Internship Day No.	2
Date	02-06-2026	Track (1 / 2 / 3 / 4)	2
Project Title	Loan Default Risk Prediction-Finance	Mentor Name	Jayasri Mam

SECTION 2 — WHAT DID YOU LEARN TODAY?

Topic / Concept Learned (e.g. Random Forest, Flask routing, model saving...)

I have learned how to create Python functions, loops, and conditional statements through practical coding exercises.

Improved problem-solving skills by building programs like Grade Calculator, ATM PIN Checker, Multiplication Table Generator, and Even/Odd Sorter.

Prepared and documented the case study for the Loan Default Risk Prediction-Finance project, including dataset selection, ML approach, and web application design.

SECTION 3 — WHAT DID YOU BUILD TODAY?

Code / Output / Feature Built (e.g. Built EDA section, trained model, created Flask form...)

Built Python programs including a Grade Calculator, ATM PIN Checker, Multiplication Table Generator, and Even/Odd Number Sorter.

Developed logic using functions, loops, conditional statements, and lists to solve real-world programming problems.

Created a detailed case study for the Loan Default Risk Prediction-Finance project, covering the dataset, machine learning approach, and web application design.

SECTION 4 — PROBLEMS FACED & HOW YOU SOLVED THEM

Error / Problem (e.g. ValueError when scaling — solved by dropping nulls first)

Encountered confusion on how to start visualizing the project and process of how to build it and process it.

How I Fixed It *(Describe your solution or what help you took)*

Resolved it by researching the project requirements and viewing other projects and structuring the content systematically.

Took help from internet to search project requirements.

SECTION 5 — PLAN FOR TOMORROW

What will you do tomorrow? *(e.g. Train Random Forest, compare with Logistic Regression, start Flask app)*

Planning to add (if possible) features that not exist in current existing solutions to cover their pitfalls. Research more about existing solutions.

Explore machine learning models for loan risk predictions.
Learn basics about NumPy & Pandas.

SECTION 6 — LINKS & ATTACHMENTS

GitHub Repo / Commit Link	GitHub Repo Day – 2 Link : https://github.com/koteswaraoposa-pixel/Innolift-intership.git
Model Result / Colab Link	
Deployed App Link (if any)	Project Case Study : https://1drv.ms/w/c/806B420EB8772E7B/IQA8dhidqLjFQqOrQJvva6uuARj-TyF3yT-VsBSjzW1wM-s?e=LaPi6l
Additional Notes	

Model Results :

```
PS C:\Users\HP\OneDrive\Desktop\innolift intership> & 'c:\Program Files\Python311\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '55487' '--' 'c:\Users\HP\OneDrive\Desktop\innolift intership\grade_calculator.py'
Enter marks for Student 1: 90
Student 1 Grade: A
Enter marks for Student 2: 65
Student 2 Grade: C
Enter marks for Student 3: 55
Student 3 Grade: D
Enter marks for Student 4: 43
Student 4 Grade: F
Enter marks for Student 5: 35
Student 5 Grade: F
PS C:\Users\HP\OneDrive\Desktop\innolift intership>
```

```
PS C:\Users\HP\OneDrive\Desktop\innolift intership> c:; cd 'c:\Users\HP\OneDrive\Desktop\innolift intership'; & 'c:\Program Files\Python311\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '50193' '--' 'c:\Users\HP\OneDrive\Desktop\innolift intership\multiplication table generator.py'
```

Table of 5

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

```
PS C:\Users\HP\OneDrive\Desktop\innolift intership> 
```

```
PS C:\Users\HP\OneDrive\Desktop\innolift intership> c:; cd 'c:\Users\HP\OneDrive\Desktop\innolift intership'; & 'c:\Program Files\Python311\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '51966' '--' 'c:\Users\HP\OneDrive\Desktop\innolift intership\ATM PIN Checker.py'
```

Enter your PIN: 1569

Wrong PIN

Enter your PIN: 3004

Wrong PIN

Enter your PIN: 1011

Wrong PIN

Card Blocked

```
PS C:\Users\HP\OneDrive\Desktop\innolift intership> 
```

```
PS C:\Users\HP\OneDrive\Desktop\innolift intership> c:; cd 'c:\Users\HP\OneDrive\Desktop\innolift intership'; & 'c:\Program Files\Python311\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '52192' '--' 'c:\Users\HP\OneDrive\Desktop\innolift intership\Function to separate even and odd numb.py'
```

Even Numbers: [12, 18, 30, 56, 22]

Odd Numbers: [7, 25, 9, 41, 13]

```
PS C:\Users\HP\OneDrive\Desktop\innolift intership> 
```

SECTION 7 — SELF RATING FOR TODAY

Area	1	2	3	4	5
Understanding of today's concept	☐ 1	☐ 2	☐ 3	☒ 4	☐ 5
Hands-on coding confidence	☐ 1	☐ 2	☐ 3	☒ 4	☐ 5
Progress on my project	☐ 1	☐ 2	☐ 3	☒ 4	☐ 5
Overall satisfaction with today	☐ 1	☐ 2	☐ 3	☐ 4	☒ 5

Case Study: Loan Default Risk Prediction Using Machine Learning

1. Project Title

Loan Default Risk Prediction System

2. Problem Statement

Financial institutions face significant losses when borrowers fail to repay loans. Traditional manual evaluation methods are time-consuming and may not accurately identify high-risk applicants. The objective of this project is to develop a machine learning-based system that predicts the probability of loan default before approving a loan application.

The system helps banks and financial organizations make informed lending decisions by analyzing customer information and estimating default risk.

3. End Users

The primary end users of this system are:

- Banks and financial institutions
- Loan officers
- Credit analysts
- FinTech companies
- Microfinance organizations

These users can evaluate loan applications quickly and reduce financial risk.

4. Dataset Selection

Dataset:

Home Credit Default Risk Dataset

Dataset Link:

<https://www.kaggle.com/competitions/home-credit-default-risk/data>

Why This Dataset?

- Contains real-world loan applicant information.
- Includes demographic, employment, income, and loan-related features.
- Large dataset suitable for machine learning training.
- Widely used for credit risk prediction research.

5. Dataset Collection Method

The dataset will be downloaded from Kaggle.

Steps:

1. Create/Login to a Kaggle account.
2. Visit the Home Credit Default Risk competition page.
3. Download the dataset files.
4. Extract CSV files.
5. Load the data using Python Pandas.

6. Data Preprocessing

The following preprocessing steps will be performed:

- Load the Home Credit dataset.
- Handle missing/null values.
- Remove duplicate records.
- Encode categorical features such as:
 - Employment Type
 - Loan Purpose
 - Education Level
- Normalize numerical features if required.
- Split data into training and testing sets.

7. Machine Learning Approach

Algorithm:

Gradient Boosting Classifier

Reason for Selection:

- High prediction accuracy.
- Handles complex relationships between variables.
- Works well with structured tabular data.
- Provides feature importance analysis.

Evaluation Metric:

ROC-AUC Score

ROC-AUC measures the model's ability to distinguish between defaulters and non-defaulters.

8. Model Development Process

Module 1 (Data Preparation)

- Load Home Credit dataset.
- Handle null values.
- Encode employment and loan-related features.

Module 2 (Model Training)

- Train Gradient Boosting model.
- Evaluate using ROC-AUC.
- Generate feature importance chart.
- Save trained model as:
loan_default_model.pkl

Module 3 (Web Application)

Develop a Flask-based web application where users enter:

- Annual Income
- Loan Amount
- Employment Information
- Loan Purpose
- Credit-related details

Output:

- Default Risk Percentage
- Loan Status:
 - Approved
 - Under Review
 - Rejected
- Top 3 Risk Factors affecting prediction

Module 4 (Deployment)

- Deploy application online.
- Generate live URL.
- Demonstrate working system.

9. Technologies Used

Programming Language

- Python

Machine Learning

- Gradient Boosting Classifier
- Scikit-Learn
- Pandas
- NumPy

Web Development

- Flask
- HTML
- CSS
- JavaScript

Database

- SQLite

Deployment

- Render / Railway / PythonAnywhere

10. Expected Output

The system predicts the probability of loan default and assists financial institutions in making safer lending decisions.

Example Output:

Applicant Income: ₹50,000

Loan Amount: ₹2,00,000

Default Risk: 18%

Status: Approved
